

NGUYEN THI THUY LINH

Trento, Italy | linhntt.unitn@gmail.com | 344-626-4074 | linkedin.com/linhntt | github.com/linhntt

Summary

Backend developer and researcher specializing in cloud computing, microservices, and security. Passionate about designing scalable, secure distributed systems from scratch. Expert in cloud-native security, RESTful API design, and container orchestration. Bridge research and industry to advance secure and resilient architectures.

Education

Soongsil University, South Korea, MS in Information & Telecommunication Mar 2022 – Feb 2024

- **GPA:** 4.28/4.5 (Transcript)
- **Thesis:** A Study on Attack Surface Reduction Through Refining Seccomp Security Profile for Containers

Hanoi University of Science and Technology (HUST), Vietnam, BS in Electronic & Telecommunication Sept 2016 – Aug 2021

- **GPA:** 3.72/4.0 (Transcript)
- **Thesis:** Data Hiding in Bit-plane Medical Image Using Chaos-based Steganography

Experience

Junior Researcher, University of Trento – Trento, Italy Jan 2025 – Present

- Collect, set up, and deploy CTF challenges on Kubernetes for UniTrento students.
- Survey in microservice security (attacks, data protection, eBPF, AI-based solutions) from 2019 to 2025
- Research about using blockchain to obtain verifiable computation in microservices systems with ZKPs.

R&D Engineer, Softfreak – Seoul, South Korea Jan 2024 – Jan 2025

- Started the API Management Security project from scratch and successfully integrated it with MyData, a financial service in the Republic of Korea [Github].
- Designed the architecture and core logic for the backend and frontend systems to interact via API.
- Developed 3 data collection agents (Nginx, eBPF, and mirroring) and 4 analysis engines integrated with Kafka and databases setup.
- Created a test case list to benchmark the system with 10,000 concurrent users using JMeter and Postman.

Research Assistant, Soongsil University – Seoul, South Korea Mar 2022 – Dec 2023

- Researched Host-Based Intrusion Detection Systems (HIDS) in container environments, utilizing AI-based approaches (LSTM models) for attack detection with system call datasets. **Publish research paper at the Mobisec 2023 conference [1].**
- Developed a prototype for Seccomp Profile generation by combining static and dynamic analysis to reduce the attack surface of containerized applications. **Publish research paper at the Mobisec 2022 conference [2].**
- Applied data balancing techniques to improve the performance of Federated Learning (FL) models in heterogeneous environments on public FL datasets (e.g., CIFAR-10, FEMNIST). **Publish research paper at the CUTE-CSA 2022 conference [3].**

Backend Engineer, Ginnovations – Hanoi, Vietnam Aug 2018 – Feb 2022

- Developed MQTT security patch for a notification system using Mosquitto, ensuring for 10,000 concurrent users.
- Designed and developed the scalable E2E protocol (inspired by Signal protocol opensource) and the Android, iOS app for XOR phones.
- Built a monitoring system for XOR phones using Logstash, Kibana, and ELK Stacks.
- Designed and implemented a C library for a secure messaging protocol using the Hybrid Encryption Scheme (ECIES) for XOR Phone.
- Built a mobile app to sync contacts on non-smartphones using audio signals and evaluating performance under XOR phones.

Research Assistant, HUST – Hanoi, Vietnam

Aug 2018 – Feb 2022

- Researched digital signal processing theory through coursework using the OMAP L138 Kit.
- Developed a novel approach for hiding information in medical images using chaos-based steganography (thesis research) and published a full paper at the IEEE MAPR 2021 Conference [4].

Publications

[1] Improving Host-based Intrusion Detection by Enhanced Contextual Information in Containers

The 7th International Symposium on Mobile Internet Security, Dec, 2023

Linh Nguyen-Thuy, Long Nguyen-Vu, Souhwan Jung

[2] Refining Seccomp Security Profile for Container Hardening

The 6th International Symposium on Mobile Internet Security, Dec, 2022

Linh Nguyen-Thuy, Long Nguyen-Vu, Jungsoo Park, Souhwan Jung

[3] TL-SMOTE: Re-balancing Data in Federated Learning for Anomaly Detection

The 14th International Conference on Computer Science and its Applications (CSA), Dec, 2022

Linh Nguyen-Thuy, Long Nguyen-Vu, Jungsoo Park, Kihoon Hong, Souhwan Jung

[4] Data Hiding in Bit-plane Medical Image Using Chaos-based Steganography

2021 International Conference on Multimedia Analysis and Pattern Recognition (MAPR), Oct, 2021

Ta Thi Kim Hue, *Nguyen Thi Thuy Linh*, Minh Nguyen-Duc, Thang Manh Hoang

Certifications

- **Postman API Fundamentals Student Expert** – Issued by Postman, Apr 2025, Credential ID
- **Implement Cloud Security Fundamentals on Google Cloud** – Issued by Google Cloud, Feb 2025, Credential ID
- **Build Infrastructure with Terraform on Google Cloud** – Issued by Google Cloud, Oct 2024, Credential ID
- **Manage Kubernetes in Google Cloud** – Issued by Google Cloud, Oct 2024, Credential ID

Awards

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| • <i>Research Fellowship, University of Trento (Duration: 1 year)</i> | 2025 |
| • <i>Master Scholarship, Soongsil University (Duration: 2 years)</i> | 2022 |
| • <i>Honda Scholarship for Science Students</i> | 2021 |
| • <i>5-Good Student Award at Hanoi University of Science and Technology (HUST)</i> | 2021 |
| • <i>2nd Prize in Student Scientific Research Competition at HUST, Vietnam</i> | 2019 |
| • <i>Scholarship with the Highest Ranking for Talented Students at University</i> | 2016 – 2021 |

Skills

- **Languages:** Kotlin, Golang, Python, Java, Android, C/C++, Matlab, Swift
- **Container Orchestration & Microservices:** Docker, Kubernetes, REST APIs
- **Database:** PostgreSQL, SQL, Redis, MariaDB, InfluxDB
- **Brokers:** RabbitMQ, Mosquitto, Active MQ, Kafka
- **Security tools:** eBPF, Sysdig, Seccomp, AppArmor, Encryption, Steganography
- **Platform:** Linux, Android, iOS, AWS
- **Monitoring:** eBPF, strace, Sysdig, ELK Stack (Logstash, Kibana)
- **Benchmarking tools:** JMeter, A/B Testing
- **Project management tools:** Jira, Agile Tools (Master Scrum), Gitlab
- **Soft skills:** Time management, Continuous Learning, Adaptability, Critical Thinking